N-of-1 Longitudinal Study of a Patient with ME/CFS: Methods and Preliminary Results

Paolo Maccallini[1]

[1]"Independent researcher", Rome, Italy

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Abstract

Keywords: n-of-1; intensive longitudinal data; ME/CFS; fNIRS; EEG; HRV; POTS

1 Introduction

2 Methods

2.1 Study design

2.2 Data acquisition

2.2.1 Symptom scores

2.2.2 Physical activity

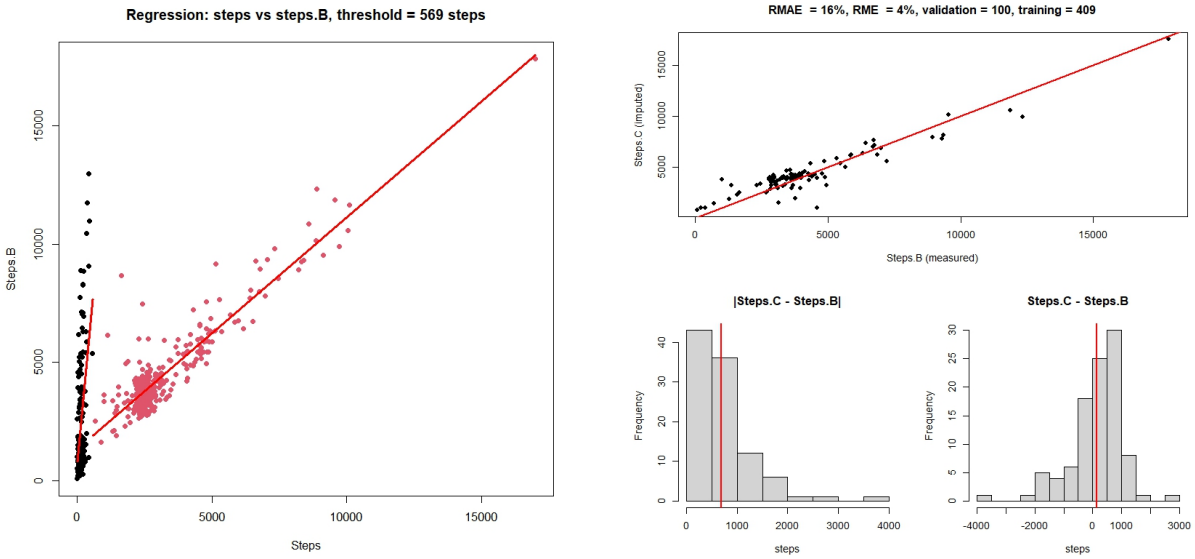
Daily step counts were recorded with a smartphone (Redmi Note 11S/Pedometer Step Counter) from January 2023, and with a wrist-worn watch (Dyurawa) from March 2025. The phone records only the steps taken while it is carried, so it misses most activity at home, whereas the watch records activity throughout the day; watch counts were therefore taken as the reference measure.

Gaps of up to four consecutive days were filled by linear interpolation in both series, if missing days were less than 10%. Days on which the watch recorded fewer steps than the phone were considered days of watch non-wear or malfunction, and their watch counts were set to missing.

To obtain a continuous series of watch-equivalent steps over the whole study period (`Steps.C`), watch counts were estimated from phone counts on days without watch data. The relationship between the two counts differs between days spent at home, when the phone is mostly not carried and its count carries little information, and days with outdoor activity, when the phone is carried. Days were therefore divided into two regimes by a threshold t on phone steps (`Steps`), and a linear regression of watch steps (`Steps.B`) on phone steps was fitted separately in each regime. The threshold was chosen by scanning values from 50 to 2000 phone steps and selecting the one that minimised the total sum of squared residuals of the two regressions, requiring at least 20

days in each regime (see Figure 1a). On days with watch data, `Steps.C` equals the measured watch count; on the other days, it is the value predicted by the regression of the corresponding regime.

The imputation was validated by fitting the model on the first 80% of the watch period and comparing the values predicted for the remaining 20% with the measured ones (Figure 1b). The relative mean absolute error, $\sum |\hat{y}_i - y_i| / \sum y_i$, and the relative mean error (bias), $\sum(\hat{y}_i - y_i) / \sum y_i$, are indicated on the plot. Most days were slightly overestimated, while the most active days, and outdoor activity without the phone, were underestimated. The time series of `Steps`, `Steps.B`, and `Steps.C` are shown in Figure 2.



(a) Regression model.

(b) Validation.

Figure 1: Imputation of watch steps (`Steps.B`) from phone steps (`Steps`).

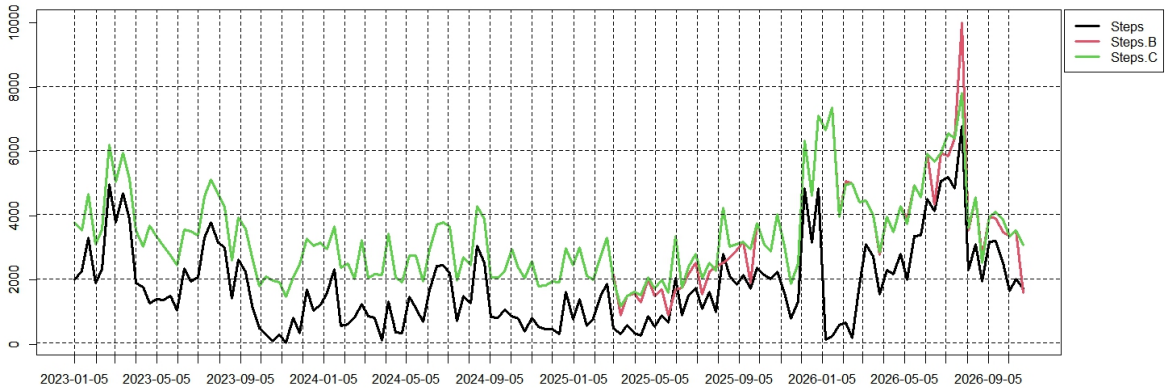


Figure 2: Daily steps averaged over non-overlapping 10-day blocks.

2.2.3 Heart rate variability

Heart rate variability (HRV) was assessed from five-minute recordings of RR intervals, following the protocol and the analysis pipeline described in detail in a previous work [1], to which the reader is referred for an overview on HRV, the definition of the indices, their physiological interpretation, and an introduction to spectral analysis by Discrete Fourier Transform. Recordings started in July 2022 and were performed once a day, in the morning, in the supine position, after five minutes of rest in the same position. RR intervals were acquired for about five minutes with a Polar H9 sensor mounted on a moistened chest belt, a type of device whose agreement with the electrocardiogram has been documented [2]. The sensor was connected via Bluetooth to a smartphone running the Elite HRV application, and the RR series were exported in text format, one file per recording.

The series were analysed with the R package RHRV [3]. For each recording, the instantaneous non-interpolated heart rate was computed from the RR intervals, and anomalous beats (ectopic beats and artefacts) were removed with the adaptive filter of the package, which discards beats that differ too much from the preceding ones or that fall outside physiological limits (default parameters). The filtered series was then linearly interpolated at 4 Hz to obtain the evenly sampled signal required by the discrete Fourier transform. Recordings shorter than 240 s or longer than 360 s were excluded, and when more than one recording was available for the same day only the first one was retained.

In the time domain, the following indices were computed: mean heart rate (HR) and mean RR interval (RR); the standard deviation of the RR intervals (SDNN) and the HRV triangular index (HRVi), as measures of overall variability; the root mean square of successive differences (rMSSD) and the percentage of successive differences greater than 50 ms (pNN50), as measures of the high-frequency, vagally mediated component [3, 4].

In the frequency domain, the power spectrum was estimated with the short-time Fourier transform, using a single window of 300 s, since for five-minute recordings shorter windows give less reliable estimates of the low-frequency power [2]. The power was integrated over the low-frequency (LF, 0.04 Hz to 0.15 Hz) and the high-frequency (HF, 0.15 Hz to 0.40 Hz) bands [5]. The ultra-low- and very-low-frequency bands were not considered, since they cannot be reliably estimated from five-minute recordings [5]. From LF and HF, the following derived quantities were computed: the ratio LF/HF; the total power, defined here as $\mathrm{TotPower} = \mathrm{LF} + \mathrm{HF}$; and the normalised powers $\mathrm{LF_{nu}} = \mathrm{LF}/(\mathrm{LF} + \mathrm{HF})$ and $\mathrm{HF_{nu}} = \mathrm{HF}/(\mathrm{LF} + \mathrm{HF})$, which reduce the within- and between-subject variability of the absolute powers [6]. The distributions of the time-domain and frequency-domain indices over all the recordings are shown in Figures 3 and 4.

2.2.4 Blood pressure and lean test

Blood pressure and heart rate were measured with an automatic upper-arm oscillometric monitor (Omron M6 Comfort) immediately after the HRV recording (Section **??**), with the subject still in the supine position, that is after at least ten minutes of supine rest. This measurement provided the supine values of Systolic (SBP) and Diastolic (DBP) blood pressure and of heart rate (HR), and was available on most days of the study.

On some days, the supine measurement was followed by a NASA lean test, performed

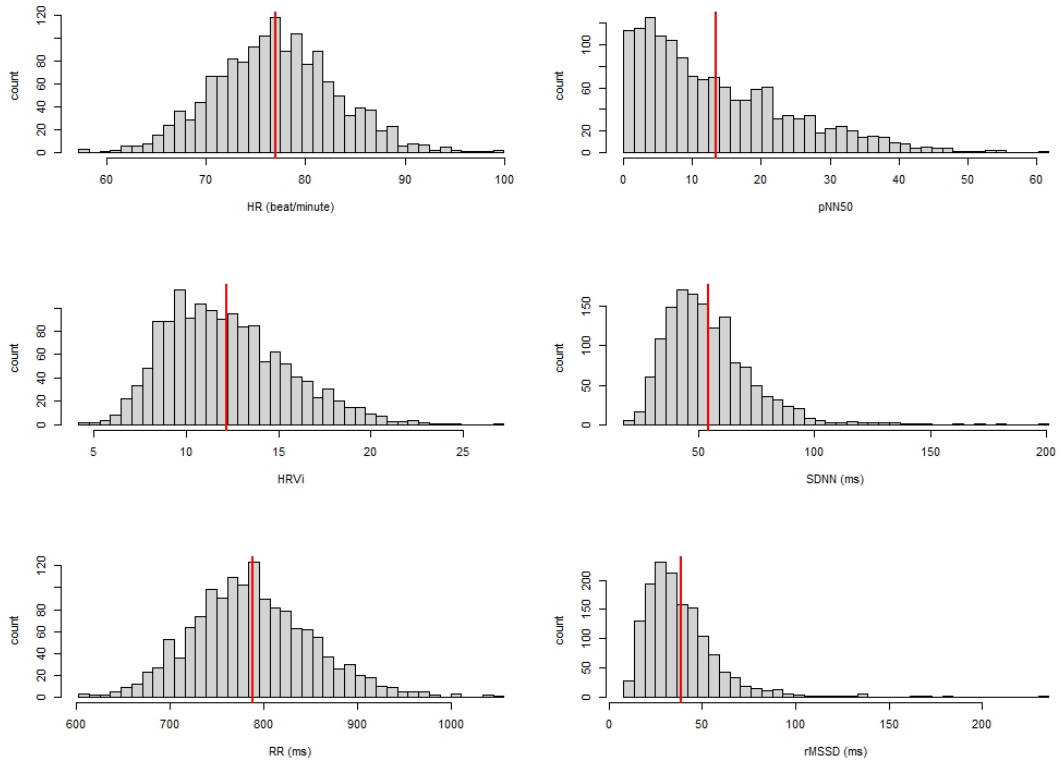


Figure 3: Distributions of the time-domain HRV indices over all the daily five-minute recordings. The mean of each distribution is indicated by a red line.

according to the protocol of the Bateman Horne Center [7], a variant of the passive standing test used by NASA to assess orthostatic intolerance [8]. The subject stood up and leaned against a wall with only the shoulder blades in contact with it and the heels about 15 cm from the wall, keeping still and without talking, a position that reduces the contribution of the leg muscle pump to venous return. Blood pressure and heart rate were then measured during standing, and each measurement was assigned to the minute of standing in which it was taken ($t = 1, 2, \ldots, 10$ min). Unlike the original protocol, which prescribes ten minutes of standing with measurements every minute, the test was performed occasionally, its duration varied between 4 and 10 minutes, and not all time points were recorded in every test; consequently, the later time points are available for fewer tests than the earlier ones. Moreover, the supine values were obtained from a single measurement rather than from two consecutive consistent readings.

For each standing measurement, the changes with respect to the supine values were computed as

$$\Delta\mathrm{HR}_t = \mathrm{HR}_t - \mathrm{HR}_0, \qquad \Delta\mathrm{SBP}_t = \mathrm{SBP}_t - \mathrm{SBP}_0, \qquad \Delta\mathrm{DBP}_t = \mathrm{DBP}_t - \mathrm{DBP}_0, \tag{1}$$

where the subscript 0 denotes the supine measurement. Pulse pressure was defined as $\mathrm{PP} = \mathrm{SBP} - \mathrm{DBP}$ and mean arterial pressure as $\mathrm{MAP} = \mathrm{DBP} + (\mathrm{SBP} - \mathrm{DBP})/3$. Each test was then summarised by the following daily indices: the maximum and the mean of ΔHR, the minimum and the mean of ΔSBP, and the minimum and the mean of PP over the available standing measurements.

Stroke volume (SV) was not measured directly. An estimate was obtained using the formula

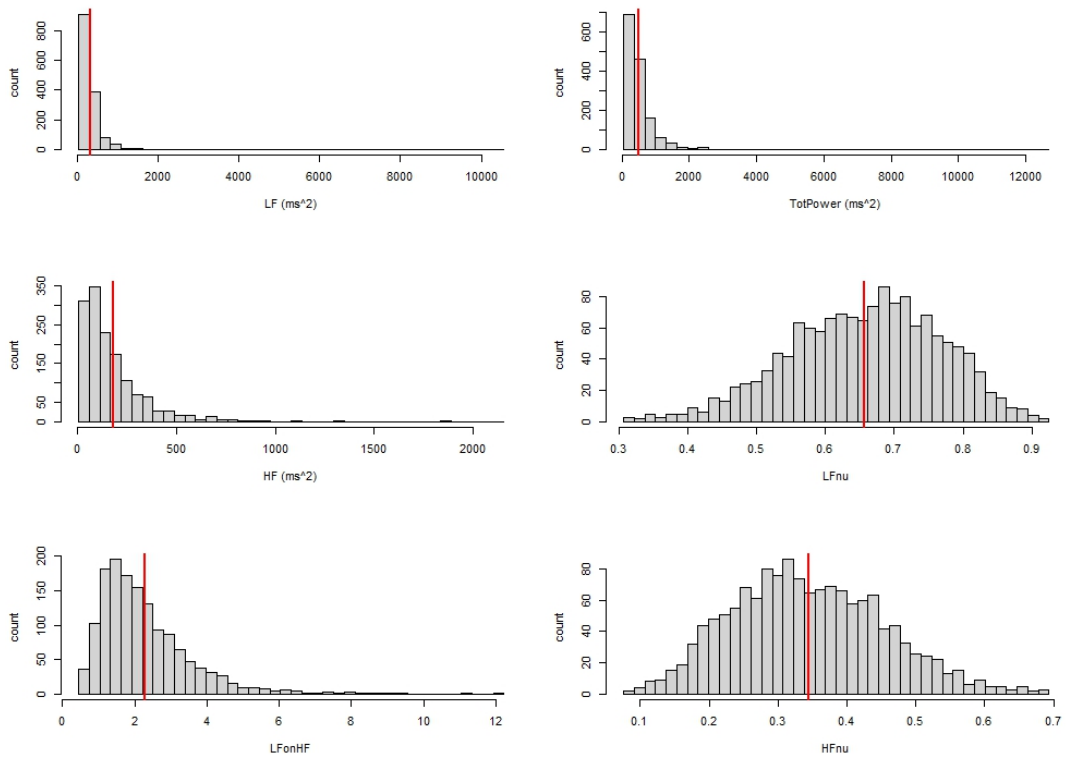


Figure 4: Distributions of the frequency-domain HRV indices over all the daily five-minute recordings. LFonHF is the ratio LF/HF; TotPower is LF + HF. The mean of each distribution is indicated by a red line.

proposed by Liljestrand and Zander [9], which expresses SV as a function of Systolic and Diastolic blood pressure:

$$\mathrm{SV} = K\frac{SBP - DBP}{SBP + DBP} \tag{2}$$

where K is a constant to be determined empirically. In a longitudinals study, like the present, the constant K has no effect on the relative changes of SV, which are the quantities of interest. Next, cardiacc output (CO) was estimated as the product of SV and HR, and cardiac power (CP) as the product of CO and MAP:

$$\mathrm{CO} = \mathrm{SV} \cdot \mathrm{HR}, \qquad \mathrm{CP} = \mathrm{CO} \cdot \mathrm{MAP} \tag{3}$$

It may be useful to remeber that SV represents the volume of blood pumped by the left ventricle in one beat, CO represents the volume of blood pumped by the left ventricle in one minute, and CP represents the mechanical power generated by the left ventricle to pump blood through the systemic circulation.

To describe the average response to standing over the whole study, each variable was plotted as a function of the minute of standing, and a second-degree polynomial was fitted to all the available measurements pooled together (Figure 5).

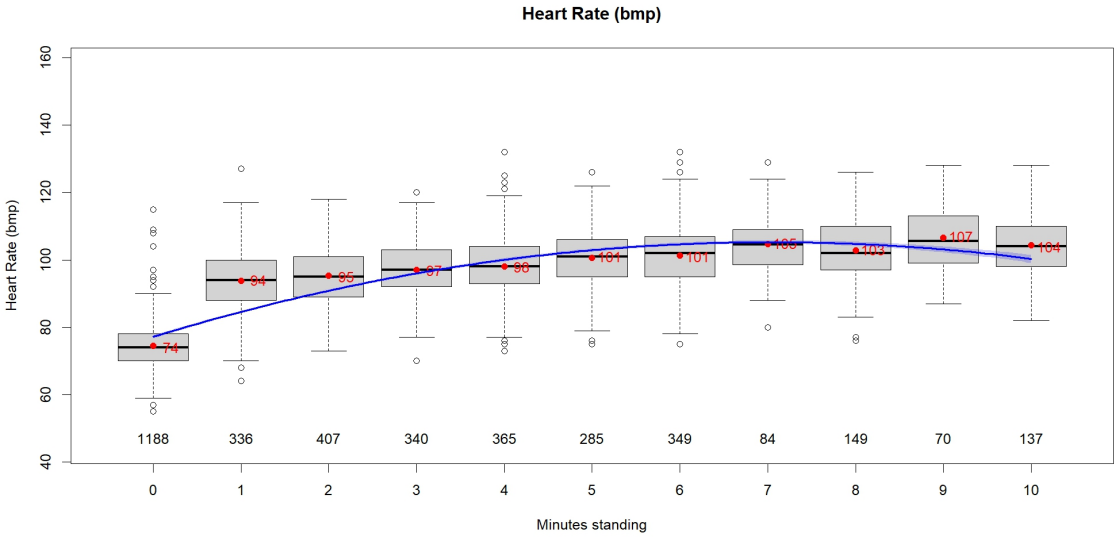


Figure 5: Heart rate during the lean test as a function of the minute of standing (0 = supine). Boxes show the distribution over all tests; red dots and numbers are the means; the numbers at the bottom are the numbers of available measurements. The blue line is a second-degree polynomial fit with its 95% confidence band.

2.2.5 Body composition

2.2.6 Indoor and outdoor environment

2.2.7 EEG and fNIRS

2.2.8 Blood tests

2.2.9 Drugs and supplements

2.3 Data processing

2.4 Statistical analysis

3 Results

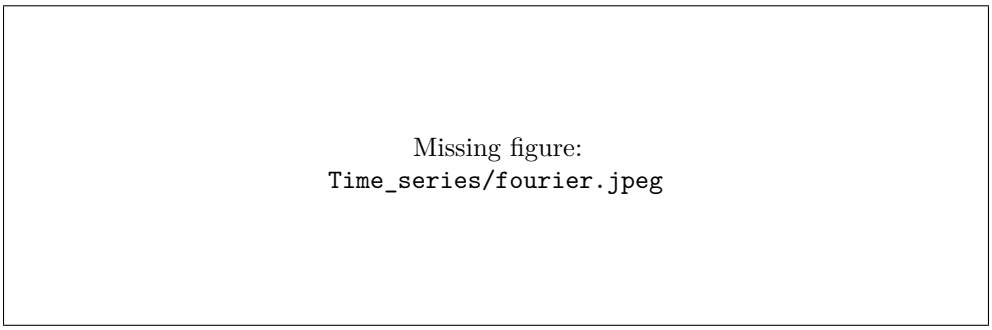


Figure 6: CAPTION.



(a) CAPTION A.



(b) CAPTION B.

Figure 7: CAPTION OF THE PAIR.

Table 1: CAPTION.

Variable	Value	Unit
NAME	0.0	ms

4 Discussion

4.1 Limitations

5 Conclusions

Data and code availability

Code: `https://github.com/USERNAME/N1-LoSt`.

Acknowledgements

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A Supplementary material